



Dermatofibrosarcoma Protuberans (DFSP): Diagnostics and Molecular Pathology

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Opinion Statement

Dermatofibrosarcoma protuberans (DFSP) is a rare, low-grade mesenchymal neoplasm that arises in the dermis and subcutaneous tissue. Clinically, DFSP presents as a slow-growing cutaneous plaque or nodular mass that is often initially mistaken for a benign dermatological condition, such as a keloid, a dermatofibroma or morphea. Due to its local aggressiveness and proclivity for subclinical spread and recurrence, accurate diagnosis and complete surgical excision are critical. Although DFSP rarely metastasises, its morbidity primarily stems from locally invasive growth and potential disfigurement due to extensive resections. The pathognomonic *COL1A1-PDGFB* fusion, which can be detected using fluorescence in situ hybridisation (FISH) or reverse transcription polymerase chain reaction (RT-PCR), sustains autocrine *PDGFRβ* activation and can be used as a diagnostic marker and a target for the drug imatinib. Emerging genomic studies have revealed additional fusions, such as *COL1A2-PDGFB* and *FBN1-CSAD*, as well as mutations, such as *CDKN2A/B* deletions, that correlate with more aggressive, fibrosarcomatous (FS) variant. We reviewed the diagnostics of DFSP, including histopathology, immunohistochemistry (e.g. CD34, factor XIIIa, S100 and PRAME) as well as clinical presentation and recommended imaging modalities (e.g. ultrasound, MRI and PET/CT). To provide a better understanding of DFSP we discussed the molecular basis of DFSP, including the main genetic drivers and downstream signalling pathways, such as *Ras-MAPK*, *PI3K-Akt* and *FGFR*. Moreover, as there is no definitive staging system for DFSP, we proposed a novel one, integrating the presence of FS differentiation.

Keywords Dermatofibrosarcoma protuberans · *COL1A1-PDGFB* gene fusion · Imatinib mesylate therapy · Differential diagnosis · Fibrosarcomatous transformation

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Introduction

Dermatofibrosarcoma protuberans (DFSP) is a rare, slow-growing soft tissue sarcoma of the skin (STS), accounting for around 1–1.8% of all STS cases [1, 2]. DFSP predominantly affects young to middle-aged adults and shows a slight male predominance with a higher incidence reported among Black individuals [1, 3]. This slow-growing mesenchymal skin tumour typically presents as indurated plaque, nodule, or multinodular cutaneous mass [4]. The term “protuberant” in its name refers to the lumps of tissue that form under the skin. Clinically, DFSP can vary from plaques or small nodular lesions to large, irregular masses. It is a locally aggressive tumour with a tendency for local recurrence, but rarely metastasises, occurring in approximately 1–4% of cases (often less than 5%) [5]. Although rare, metastasis is more likely to occur in certain circumstances. For example, it is more likely to occur if the tumour is large or located in specific areas, such as the head and neck [6]. Classic DFSP has a favourable prognosis, with a five-year progression-free survival (PFS) rate of around 93% [7]. However, the prognosis is significantly worse for fibrosarcomatous transformation (FS), a high-grade variant, with a 5-year PFS rate of just 33% and a 14.4–57% risk of metastasis; (median 6.8 years), 5-year OS was 64% (median time for OS was not reached) [7, 8]. LR occurs in up to 58% of FS-DFSP cases, and though it is uncommon in classic DFSP, distant metastasis becomes a notable concern with FS change [7]. Surgical excision remains the cornerstone of treatment. Mohs micrographic surgery (MMS) offers the lowest LR rates thanks to complete margin evaluation. Studies have shown that the recurrence rate for MMS is significantly lower (around 1–2%) than for wide local excision (WLE; 7–30%) [1, 9–11]. Yet, WLE with margins of 2–3 cm is an acceptable alternative [12]. For locally advanced or unresectable disease, imatinib, a tyrosine kinase inhibitor that targets the *COL1A1-PDGFB* fusion gene, is an effective treatment, particularly in cases where the genetic mutation has been confirmed. Objective response rates are over 50% as was shown by our clinic [7, 13, 14]. Therefore, a correct diagnosis is essential as FS differentiation affects both prognosis and the therapeutic strategy. To guide the use of targeted therapies like imatinib, molecular confirmation via FISH or RT-PCR for *COL1A1-PDGFB* fusion is recommended as stated in multiple guidelines including ESMO and NCCN [12, 15–18].

Clinical presentation

Although DFSP is usually superficial and confined to the dermis, it can invade fascia, muscle, periosteum and bone [19]. The tumours often have asymmetrical, subclinical horizontal finger-like extensions in the skin, making margin control difficult [20]. Diagnosis is often delayed because the lesions are slow-growing and may initially appear benign [21]. Some cases of DFSP do not protrude above the skin and may present as non-protuberant plaques which can last for an average of 7 years before the lesion becomes bulky and nodular [22]. The tumours can occur at any age, but are most common in patients between the ages of 20 and 50 [5, 23]. The most common site is the trunk (50–60%), followed by the extremities (20–30%) (the distribution among the extremities is often reported as fairly equal), and the head and neck (10–15%) [24, 25]. In the head and neck, the most common sites are the supraclavicular area and the forehead [26]. DFSP has also been reported in other areas such as the face, within the breast tissue, in the pubic region and vulva, and even in the oral cavity (including areas such as the cheek) [13–19]. The size of DFSP tumours at diagnosis varies, but the mean is around 2–3.5 cm [27–29]. However, tumours have been reported much larger, with a case report with DFSP on the trunk reaching nearly 40 cm in size [30]. DFSPs present as multinodular lesions [31]. In the early stages, DFSP lesions are often painless [31], but as the tumour progresses, especially in the rapidly growing variant, it can become painful and may ulcerate or bleed [32]. During pregnancy, DFSP frequently exhibits an accelerated growth rate, a phenomenon that may be attributed to the hormonal changes associated with gestation [33]. Furthermore, DFSP may be misdiagnosed as other benign conditions (Table 1), such as scars, due to its similar clinical appearance [77, 78], bruises [79], morphea [52], dermatofibroma [34], hemangioma [80] and other benign soft tissue tumours because of its slow growth and non-specific appearance. It may also be misdiagnosed as nodular fasciitis [81], fibroma [82], neurofibroma [72, 83], hematoma [84], schwannoma [85], low-grade fibrosarcoma [86], synovial sarcoma [87, 88], spindle cell melanoma [64], fibrosarcoma [89], or atypical fibroxanthoma [90]. Due to the rarity of this condition, general practitioners are not as familiar with the presentation and may misinterpret it as a benign condition, leading to delays in referral to dermatologists or oncologists. General practitioners need to be aware of slow-growing, non-painful skin lesions, especially those that persist or grow rapidly despite standard treatment, which may indicate DFSP.

Table 1 Differential diagnosis of DFSP from other diseases

Disease	Main Differences from DFSP	References
Dermatofibroma (DF)	-benign lesion -prominent epidermal hyperplasia with peripheral collagen trapping -variable cellularity (spindle cells, multinucleated giant cells, siderophages) -typically CD34 (–) or only focally positive; factor XIIIa (+) -commonly located in the upper to mid-dermis (grenz zone)	[27, 34, 35]
Cellular Dermatofibroma	- CD34 (+) -superficial biopsies can be challenging.	[36–38]
Plaque-like CD34-positive dermal fibroma	- Benign lesion - CD34 (+) - Typical location: upper mid-dermism, grenz zone	[39, 40]
Dermatomyofibroma	-Benign lesion -Exhibits a fascicular growth pattern without storiform architecture. -Lacks the honeycomb fat infiltration seen in DFSP.	[41–43]
Epidermal Inclusion Cyst	- epidermal cyst: nodule with a visible central punctum	[44, 45]
Keloid and Hypertrophic Scar	-post-traumatic, firm, raised lesions - imaging (ultrasound, dermatoscopy, MRI) aids differentiation	[46–48]
Melanoma	-S100 (+) -distinct dermatoscopic features	[49–51]
Morphea/Localised Scleroderma	-dense dermal collagen; lacks storiform pattern and honeycomb fat infiltration.	[52–55]
Lymphoma	- lymphoid marker (+).	[55–57]
Fibrosarcoma	-CD34 (–) (DFSP is CD34 (+); DFSP-FS may show weak expression) -deep soft tissues; lacks DFSP's dermal origin and honeycomb fat infiltration	[58–60]
Leiomyosarcoma	- marked atypia, high mitotic rate, fascicular growth, diffuse, no honeycomb fat infiltration. -desmin/smooth muscle marker positivity	[43, 61]
Malignant Peripheral Nerve Sheath Tumour (MPNST)	- S100 (+), CD34 (–); -typically located deeper than DFSP.	[62, 63]
Spindle Cell Melanoma	-S100 (+), melanocytic markers (+), CD34 (–); -distinct growth pattern compared to DFSP	[64, 65]
Solitary Fibrous Tumour	-CD34 (+); -haphazard growth with staghorn vessels; -lacks storiform pattern and honeycomb fat infiltration	[66, 67]
Myxoid Liposarcoma	- CD34 (–) - may exhibit lipoblasts (absent in myxoid DFSP, which is CD34 (+))	[43, 68]
Low-grade Myxofibrosarcoma	- multilobular pattern, curvilinear vessels, and atypia.	[43, 69]
Superficial Angiomyxoma	-less cellular; lobular growth pattern; no honeycomb fat infiltration; -CD34 (+)	[70, 71]
Neurofibroma	- S100 (+); -wavy, buckled nuclei	[27, 72]
Nodular Fasciitis	-infiltrative; mucinous stroma; multinucleated giant cells; frequent lymphocytes	[73, 74]
Granular cell tumour	- S100 (+), CD63 (+) - granular cytoplasm	[56, 75, 76]

Diagnosis

In DFSP, ultrasound is used for initial assessment, preoperative planning and, in some cases, postoperative monitoring for recurrence [91, 92]. DFSP typically presents as a hypoechoic or heterogeneously hypoechoic mass with well-defined margins. The lesion often infiltrates the subcutaneous fat, forming a honeycomb pattern characteristic of DFSP. Doppler ultrasound may show increased vascularity within the tumour, indicating its infiltrative and proliferative nature [93]. High-frequency probes (10–20 MHz) provide excellent resolution for delineating tumour depth, subcutaneous extension and possible invasion into deeper structures such as muscle or fascia [94]. Ultrasound can reveal peri-tumoural oedema, which is common in DFSP, and can help identify subtle extensions beyond clinically visible margins [95].

Diago et al. [96] evaluated the use of ultrasound in the assessment of DFSP in 30 cases and demonstrated a high concordance with histopathology for measuring tumour depth. Tumours appeared predominantly as oval, hypoechoic lesions, with finger-like projections identified in 73.3% of cases and a posterior hyperechoic area corresponding to honeycomb invasion in 53.3%. Ultrasound depth measurements (mean 7.3 ± 4.0 mm) agreed closely with histopathological depths (mean 8.3 ± 4.2 mm), with an intraclass correlation coefficient of 0.97. The sensitivity for detecting muscle or fascia invasion was 81.8% and the specificity was 100%, highlighting the value of ultrasound in the preoperative planning and surgical management of DFSP. In another study [97], recurrent DFSP were more likely to be located in deeper structures (62.9%) and presented as oval, lobulated or irregular masses. Both groups had hypervascularity on Doppler imaging. Ultrasound showed a diagnostic accuracy of 88.6% for recurrent DFSP but only 17.4% for primary DFSP.

MRI, on the other hand, is more suitable for preoperative assessment, especially for larger or recurrent tumours (Table 2). It provides a precise understanding of tumour depth and lateral spread, with a sensitivity of 67% and specificity of 100% [91]. On T1-weighted images, DFSP lesions typically appear hypointense relative to subcutaneous fat, while on fat-saturated T2-weighted sequences, tumours show slightly heterogeneous hyperintense signals with patchy hypointense areas. These features are critical for distinguishing DFSP from surrounding tissue [95]. Postoperatively, MRI is highly effective in identifying recurrences, showing nodular, homogeneous, well-defined lesions with marked contrast enhancement (Fig. 1) [95]. analysed the imaging findings in 27 patients with DFSP, including 18 cases with CT and 9 cases with MRI. Tumours ranged from 2 to 17 cm in diameter, with the majority being

well-circumscribed, solitary, and infiltrating subcutaneous fat without calcification. Contrast-enhanced CT scans showed moderate to marked heterogeneous enhancement in 48.1% of cases and moderate homogeneous enhancement in 14.8%. On MRI, T1-weighted images showed isointense lesions in 9 cases, while T2-weighted images showed hyperintense or mixed signal intensities. DFSP was characterised histopathologically by spindle cells in storiform arrangement, with CD34 positivity observed in all cases. There were no metastases at diagnosis and 92.6% of cases showed no recurrence during a follow-up period of 6 months to 3 years.

CT imaging is generally reserved for specific cases, such as suspected bone invasion or pulmonary metastases [103]. For example, CT has demonstrated a positive predictive value (PPV) of 100% for identifying periosteal margin positivity in cases of DFSP located on the scalp, although its sensitivity is lower at 25% [109]. CT is beneficial in preoperative planning for large or complex tumours, providing clear anatomical details to guide surgical strategies [110].

Although not a routine diagnostic tool for DFSP, PET scans have demonstrated utility in specific scenarios, such as assessing treatment response or detecting metastases. For example, in one case where FDG-PET imaging was used to monitor response to imatinib therapy, tumour FDG uptake decreased significantly within two weeks, correlating with clinical improvement [105]. PET/CT has also been shown to be effective in delineating disease extent, particularly in high-grade or fibrosarcomatous DFSP, where aggressive metabolic activity can be identified [111]. The diagnosis of DFSP is not always straightforward and is often delayed. Clinical suspicion based on the appearance of the lesion is usually followed by biopsy for confirmation [112]. Incisional or excisional biopsy is preferred, whereas fine-needle aspiration is usually avoided due to its relative inaccuracy [113]. Diagnosis requires histopathological analysis, and the use of immunohistochemical stains may be helpful. DFSP shows a diffuse infiltration of the dermis and subcutaneous tissue by spindle-shaped cells in a storiform pattern together with tumour cells, typically spreading along the septa of the subcutaneous adipose tissue [114]. The tumour cells are usually CD34 positive [115]. Molecular tests such as fluorescence in situ hybridisation (FISH) or multiplex reverse transcriptase polymerase chain reaction (RT-PCR) are useful in confirming difficult diagnoses of DFSP by detecting specific chromosomal translocations and fusion gene transcripts [116, 117]. These tests can confirm the presence of a COL1A1-PDGFB fusion, which is characteristic of DFSP [118].

There is no definitive staging system for DFSP. Some studies use the American Joint Committee on Cancer

Table 2 Different non-invasive imaging methods used in DFSP

Diagnostic Method	Description	Advantages	Limitations	Applications	References
Ultrasound (US)	-Widely available imaging modality. -Detects hypoechoic masses in DFSP. -Identifies pseudopodia-like projections. -Assesses tumour boundaries with sensitivity 81.8% and specificity 100%	-Real-time imaging, cost-effective and useful for initial assessment of tumour depth and extent. -Helps to differentiate DFSP from other dermal tumours when pseudopod-like projections are observed. -Non-invasive and suitable for repeat imaging.	-Highly operator dependent, with interpretation dependent on skill and experience. -Imaging features are not entirely specific and may overlap with other benign or malignant lesions.	Useful in initial assessment and preoperative planning to assess tumour size and depth.	[77, 96, 97]
High-Frequency Ultrasound (HFUS)	-Higher resolution than standard ultrasound. -Particularly effective for vascular imaging and margin assessment.	-Excellent for delineating tumour boundaries. -Suitable for the detection of sub-clinical lesions. -Can guide biopsies for histopathological or molecular studies.	-Requires specialised equipment and experienced operators. -Limited use in routine practice due to availability.	Best suited for differential diagnosis of vascular lesions and preoperative margin assessment.	[98, 99]
Computed Tomography (CT)	-Essential for assessing advanced or recurrent DFSP. -Evaluates tumour invasion into bone and adjacent structures. -Identifies pulmonary metastases.	-Provides detailed imaging of bone involvement in aggressive or long-standing DFSP. -Allows comprehensive staging in recurrent or metastatic cases.	-Exposure to ionising radiation. -Limited differentiation of soft tissue structures without contrast enhancement	Recommended for advanced staging in cases where bone involvement or metastases are suspected.	[100–102]
Magnetic Resonance Imaging (MRI)	-High-resolution soft tissue imaging. -Detailed visualisation of tumour depth and infiltration.	-Superior for assessment of tumour extension into subcutaneous fat, fascia and muscle. -High sensitivity (67%) and specificity (100%) for margin assessment compared to palpation. -Non-invasive and radiation free.	-Costly and less accessible than ultrasound. -May have limited effectiveness in certain anatomical regions, such as the head and neck.	Ideal for preoperative planning, recurrence monitoring and evaluation of complex cases.	[100, 103, 104]
High-Resolution MRI (HR-MRI)	-Improved visualisation of microcirculation. -Enhanced detection of tumour infiltration.	-Accurate in defining complex or deeply infiltrated cases.	-Requires advanced imaging infrastructure and training.	Complex case evaluation during preoperative planning.	[74]
Positron Emission Tomography-CT (PET-CT)	-Combines functional (metabolic) imaging with anatomical detail. -Aids in staging and assessing treatment response.	-Tracks tumour activity and response to therapies such as imatinib. -Detects distant metastases and recurrence.	-Expensive and not widely used for routine DFSP diagnosis. -High radiation exposure.	Reserved for advanced or recurrent disease, especially at metastatic stage.	[100, 105]
Optical Coherence Tomography (OCT)	-Produces cross-sectional skin images. -Visualises architectural abnormalities and skin thickening in DFSP.	-Non-invasive, with real-time imaging capabilities. -Helps to identify margins in tissue-sparing surgery. -Potentially improves diagnostic accuracy in equivocal cases.	-Limited penetration. -Requires operator skill and specialised equipment.	Most effective for preoperative surgical margin assessment and follow-up.	[100, 106]
Dermatoscopy	-Evaluates superficial features, including vascular patterns and pigmentation. -Differentiate DFSP from benign lesions.	-Widely available and cost effective. -Useful for initial lesion characterisation.	-Lacks specificity and cannot assess tumour depth. -Best used in combination with other imaging modalities.	Used during initial screening to identify suspicious lesions requiring further investigation.	[100, 107, 108]

(AJCC) staging protocol for soft tissue sarcomas, or a simple distinction between primary tumour and local recurrence. The American Musculoskeletal Tumour Society (MSTS) system is based on histological grade, local extension and

the presence of metastases, and because DFSP is almost always low-grade, it is usually staged as IA (low-grade tumour confined to an anatomical compartment) or IB (low-grade tumour not confined to an anatomical compartment).

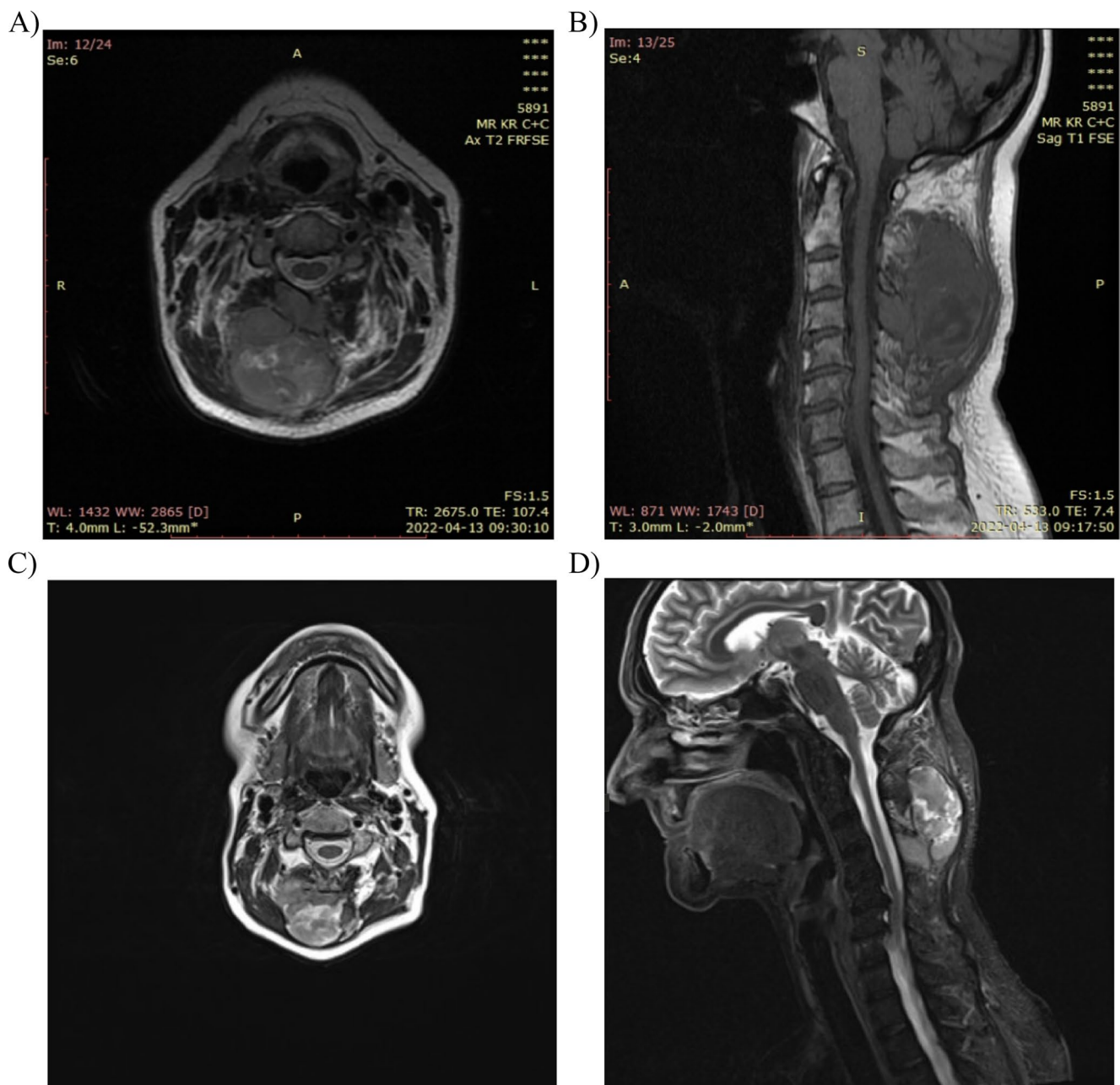


Fig. 1 A 61-year-old female patient treated for dermatofibrosarcoma protuberans of the neck with fibrosarcomatous transformation. The tumour was located in the posterior aspect of the cervical spine and was considered unresectable at initial presentation. Contrast-enhanced

axial (**A**) and sagittal (**B**) MRI images showing preoperative tumour appearance; Contrast-enhanced axial and sagittal preoperative MRI images showing significant reduction of the tumour size post chemotherapy and post radiation (**C**, **D**).

The NCCN classification is based on local, regional and distant disease, and the 2018 German guidelines classify by primary tumour (localised disease), lymph node metastasis and distant metastasis. We propose a new staging system for DFSP present in Table 3.

Pathology

Microscopically, classic DFSP typically shows a proliferation of uniform spindle cells arranged in a storiform pattern [119]. These cells are elongated with mild hyperchromasia,

Table 3 Proposed staging system for DFSP

Stage	Tumour Characteristics (T)	Fibrosarcomatous Change (FS)	N (Lymph Node Involvement)	M (Metastasis)	Description
Stage I	T1: ≤5 cm, confined to the dermis or subcutaneous tissue	FS-: No fibrosarcomatous change	N0	M0	Localised, low-risk DFSP without FS transformation, amenable to wide local excision.
Stage IIa	T2a: >5 cm and ≤10 cm, confined to subcutaneous tissue	FS-: No fibrosarcomatous change	N0	M0	Larger localised DFSP with increased risk of local recurrence, still without FS transformation.
Stage IIb	T2b: >5 cm and ≤10 cm, with deep invasion (e.g., muscle or fascia)	FS-: No fibrosarcomatous change	N0	M0	Intermediate DFSP invading deep structures but without FS transformation.
Stage IIIa	Any size, superficial or deep	FS+: Fibrosarcomatous change present	N0	M0	DFSP with FS transformation, which carries a higher risk of local recurrence, metastasis, and mortality.
Stage IIIb	T3: Tumour >10 cm or any size with critical structure involvement (e.g., bone, nerves)	FS- or FS+	N0	M0	Advanced local disease, with or without FS transformation, requiring aggressive treatment.
Stage IV	Any size	FS- or FS+	N1: Regional lymph node involvement	M1: Distant metastasis	Metastatic DFSP, more commonly observed in FS cases, involving lungs or other distant organs.

small to inconspicuous nucleoli, and scant cytoplasm (Fig. 2) [120]. DFSP spindle cells infiltrate the dermis and subcutaneous tissue, often creating a honeycomb pattern within the subcutaneous fat [121, 122]. Immunohistochemically (IHC) tumour cells consistently express CD34, which is characteristic DFSP marker [121]. CD34 expression helps distinguish DFSP from other spindle cell neoplasms spindle cell lipoma, solitary fibrous tumour, neurofibroma, peripheral nerve sheath tumour, angiosarcoma and myxoid tumour. Although CD34 positivity is not exclusive to DFSP and can be seen in other tumours like spindle cell lipomas and solitary fibrous tumours. Other IHC markers including SMA, Desmin, Cadesmon, S100, SOX10, FXIIIa, STAT6 are used in the differential diagnosis to exclude benign lesions (i.e. dermatofibromas), low-grade sarcomas (i.e. leiomyosarcoma), melanoma (i.e. S100, SOX10, HMB45) [49] or malignant peripheral nerve sheath tumours (i.e. H3K27me) (Table 4) [120, 133].

DFSP lacks significant cell pleomorphism and high mitotic activity, which are described in high-grade sarcomas such as fibrosarcomas. Additionally, DFSP can occasionally exhibit fibrosarcomatous transformation, with increased cellularity, pleomorphism, and mitotic activity, resembling fibrosarcoma (Table 5). Other morphological DFSP variants

include pigmented (Bednar tumour), myxoid, and plaque-like DFSP, DFSP with myoid differentiation,

The tumour microenvironment in DFSP includes variable lymphocytic infiltration, though it is generally sparse compared to other malignancies. There is limited involvement of macrophages and natural killer (NK) cells, which suggests that DFSP might not be targeted by conventional immunotherapy [120, 133] such as checkpoint inhibitors like anti-PD-1/PD-L1 (e.g. pembrolizumab or nivolumab) or anti-CTLA-4 therapies (e.g. ipilimumab). These therapies work by blocking inhibitory signals on T cells, thereby reactivating immune responses against tumours. However, in DFSP, the sparse lymphocytic infiltration and limited presence of cytotoxic immune cells suggest a reduced likelihood of response to these agents [144]. Checkpoint inhibitors are most effective in tumours with a high mutational burden or robust immune infiltration, neither of which is characteristic of DFSP. This low presence of immune cells, often referred to as an ‘immune-cold’ tumour microenvironment, suggests that DFSP may require strategies aimed at converting the tumour to an ‘immune-hot’ phenotype [145].

However, the presence of the *COL1A1-PDGFB* fusion gene in DFSP cells opens the possibility for targeted therapies using tyrosine kinase inhibitors; in routine

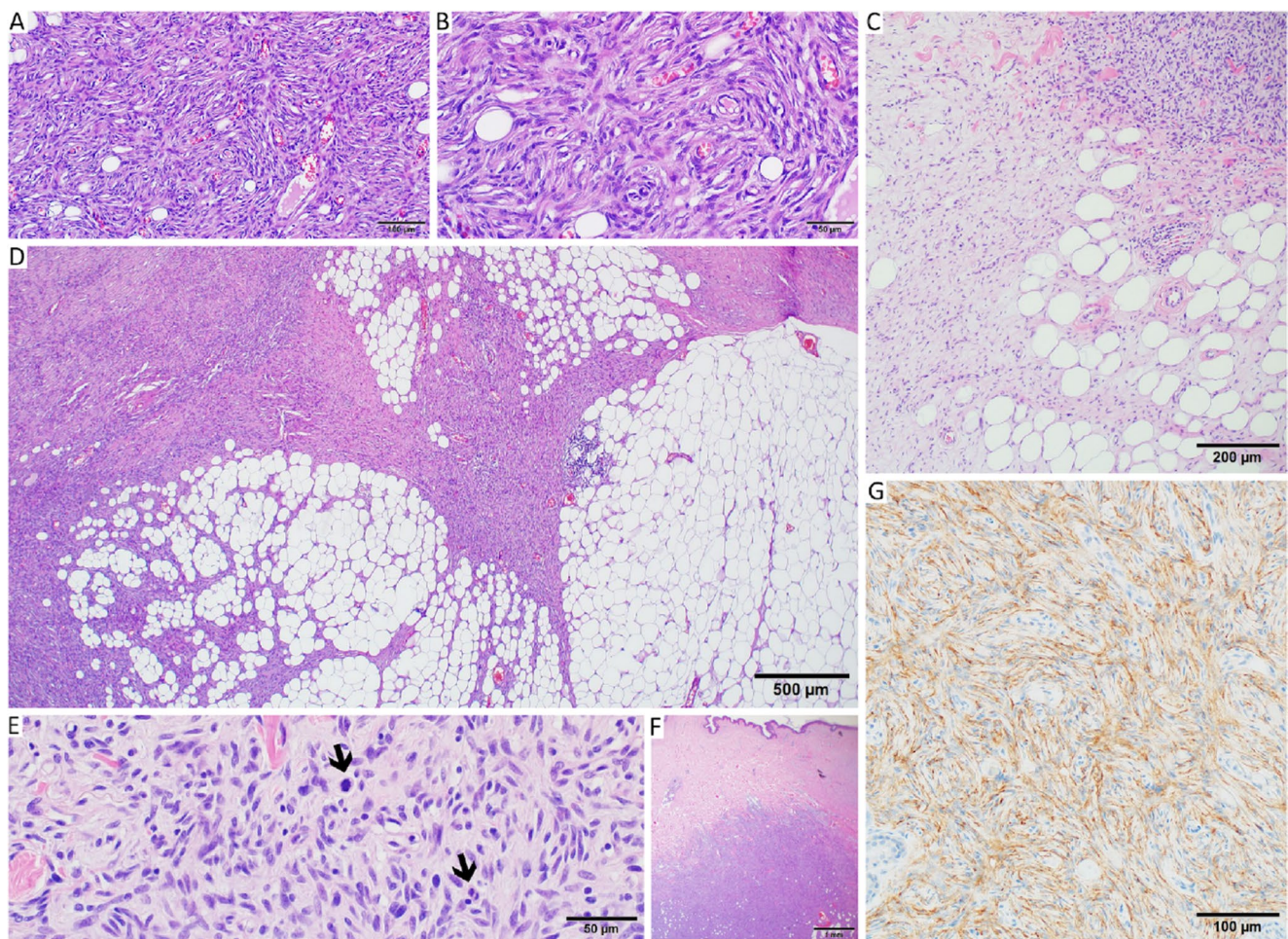


Fig. 2 Histopathological image of DFSP. (A) Spindle cells with low cytological atypia (HE); (B) storiform pattern (HE); (C) myxoid morphological variant is the most common (HE); (D) DFSP typically is arranged in honeycomb pattern infiltrating subcutaneous fat (HE);

(E) brisk mitotic activity is seen in fibrosarcomatous transformation (HE), which is (F) usually located in deeper parts of the skin and presents increased cellularity and pleomorphism (HE); (G) CD34 is highly expressed on DFSP cells.

Table 4 Immunohistochemical marker profiles of dermatofibrosarcoma protuberans and differential diagnoses

Tumour Type	Histology	CD34	Factor XIIIa	SMA	STAT6	RB	SOX10	S100	HMB45	References
Dermatofibrosarcoma	-Storiform pattern. -Honeycomb infiltration in fat.	+	-	-	-	-	-	-	-	[114, 123, 124]
Dermatofibroma	-Cellular pleomorphism. -Spindle cells and histiocytes.	-	+	-	-	-	-	-	-	[34]
Fibrosarcoma	-Herringbone pattern. -Significant atypia. -High mitotic rate.	+/-	-	-	-	-	-	-	-	[125]
Solitary Fibrous Tumour	-Patternless, spindle cells with thick collagen.	+	-	-	+	-	-	-	-	[66, 67]
Nodular Fasciitis	-Myxoid stroma. -Rapid growth, high mitotic rate.	-	-	+	-	-	-	-	-	[126, 127]
Spindle Cell Lipoma	-Bland spindle cells. -Myxoid matrix. -Ropey collagen.	+	-	-	-	-	-	-	-	[128]
Spindle Cell Melanoma	-Atypical spindle cells. -Melanocytic features.	-	-	-	-	-	+	+	+	[129–132]

Table 5 Immunohistochemical markers for variants of DFSP

DFSP variant	CD34	Factor XIIIa	S100	Desmin	PRAME	SMA	Diagnostic features	References
Conventional/Classic	Positive (strong, diffuse)	-	-	-	-	-	- CD34 (+) confirms diagnosis. -SMA (-) and S100 (-) rule out other spindle cell tumours. -Vimentin (+), apolipoprotein D (+), nestin (+). -CD34 loss: indicates higher-grade transformation. -Ki-67: highlights proliferative activity. -Vimentin (+). -PRAME (+) in 43%.	1 [114]
Fibrosarcomatous	Weak or absent, may be focally positive	-	-	-	Positive or higher H-score	Negative or focally positive in myoid areas	-Melanocytic markers differentiate from conventional DFSP. -Melan-A (-), HMB-45 (-) [136]	2 3 [134, 135]
Pigmented (Bednar Tumour)	Spindle cells positive, pigmented cells negative	-	Pigmented cells positive, spindle cells negative	-	-	-	- Myxoid stroma; retained CD34 (+). -Nodular growth pattern.	4 5 [137, 138]
Myxoid	+	-	-	-	-	-	- Thinned dermis. -CD34 (+). -Depressed plaque; parallel fascicles.	6 67 [42, 139]
Atrophic	+	-	-	-	-	-	-Granular eosinophilic cells.	44 55 [140]
Granular cell DFSP	+	-	-	-	-	-	- Myoid nodules; perivascular cells; eosinophilic cytoplasm. -SMA weak (+).	5 66 [141, 142]
Myoid DFSP	Weak or negative in myoid areas, positive in classic areas	-	-	-	-	Positive in myoid areas, patchy or focal		7 77 [42, 143]

histopathological diagnostics this fusion gene is detected in fluorescence in situ method (FISH) and is very helpful in the diagnosis confirmation [2, 4, 122, 146]. This fusion gene not only aids in diagnosis, but also serves as a critical therapeutic target for imatinib mesylate, a potent tyrosine kinase inhibitor specifically designed to block the activity of PDGFR β [147]. The COL1A1-PDGFB fusion gene leads to overproduction of platelet-derived growth factor beta (PDGFB), which drives tumour growth through an autocrine signalling mechanism that promotes uncontrolled cell proliferation and survival. Imatinib binds to the ATP-binding site of PDGFR β , preventing its activation and subsequent downstream signalling. This targeted approach is particularly beneficial in cases where surgical resection is not feasible due to tumour location, size or recurrence with positive margins. Clinical studies have shown significant tumour shrinkage and disease stabilisation in patients with unresectable or metastatic DFSP when treated with imatinib at doses of 400 to 800 mg/day (importantly [147], there seems to be no difference between 400- or 800-mg daily doses) [148]. Thus, the presence of the COL1A1-PDGFB fusion gene not only underpins the molecular pathogenesis of DFSP, but also provides a specific avenue for effective pharmacological intervention.

Accurate diagnosis of DFSP requires careful histopathological examination and an appropriate panel of immunohistochemical stainings, supplemented by molecular techniques such as FISH or reverse transcription-polymerase chain reaction (RT-PCR) to detect specific gene fusions [117].

Genetics and molecular biology

DFSP is associated with specific mutations related to chromosomes 17 and 22 which may appear as supernumerary ring chromosomes or chromosomal translocation t(17; 22) (q22; q13) (Table 6) [27, 159]. It results in a gene fusion of COL1A1 (17q21) with PDGFB (22q13) occurs (Fig. 3). As a consequence, there is observed an overexpression of PDGFR β which causes continuous activation of his receptor. A correlation exists between the aforementioned mutation and activation of the Akt-mTOR pathway [160] which appears in many sarcomas [161] and is involved in the tumorigenesis of DFSP [27]. Fluorescent in situ hybridization (FISH) or multiplex reverse transcriptase-polymerase chain reaction (RT-PCR) can be used in diagnosis to detect fusion COL1A1-PDGFB [27, 162]. Whole genome

Table 6 Genetic and molecular alterations in DFSP

Mutation/Genetic Alteration	Description	Significance in DFSP	References
<i>COL1A1-PDGFB</i> Fusion	- Most common alteration in DFSP: t(17;22)(q22;q13) translocation or supernumerary ring chromosomes. - Fusion gene: exon 2 of PDGFB with exons 6–49 of COL1A1.	- Hallmark diagnostic marker of DFSP. - Drives constitutive activation of PDGFR β signalling, leading to uncontrolled cell proliferation. - Primary target for imatinib, which blocks PDGFR β activation.	[149–151]
<i>COL1A2-PDGFB</i> Fusion	- A rare fusion variant involving COL1A2 exon 26 and PDGFB exon 2.	- Activates the PDGF signaling pathway but is significantly less prevalent. - May exhibit imatinib sensitivity	[122]
<i>FBN1-CSAD</i> Fusion	- A novel fusion identified in fibrosarcomatous DFSP. Involves FBN1 exon 42 and CSAD exon 2.	- FBN1 is linked to extracellular matrix assembly, while CSAD is involved in cysteine metabolism	[152]
<i>COL6A3-PDGFD</i> Fusion	- Rare fusion involving COL6A3 exon 42–43 and PDGFD exon 6, predominantly seen in DFSP cases in the female breast.	- Activates the PDGF pathway, contributing to tumorigenesis.	[153]
<i>EMILIN2-PDGFD</i> Fusion	- Fusion of EMILIN2 exon 4 with PDGFD exon 6.	- Leads to PDGF signaling activation.	[154, 155]
<i>SLC2A5-BTBD7</i> Fusion	- Novel fusion identified via t(1;14), involving <i>SLC2A5</i> and <i>BTBD7</i> .	- Emerging evidence suggests potential roles in tumorigenesis.	[156, 157]
<i>MUC4/MUC6</i> Mutations	- Mutations in <i>MUC4</i> or <i>MUC6</i> genes, found to be mutually exclusive.	- Occur in 25% (<i>MUC4</i>) and 21% (<i>MUC6</i>) of cases, respectively. - Used to classify DFSP into molecular subtypes for potential prognostic or therapeutic insights.	[156]
<i>TLK2</i> Amplification	- Amplification of the Tausled-Like Kinase 2 (TLK2) gene.	- Correlates with poor prognosis. - Plays a role in genome maintenance, cell division, and immune evasion.	[152]
<i>MAP3K7CL-ERG</i> Fusion	- A novel fusion detected in FS-DFSP involving MAP3K7CL and ERG genes.	- Suggests additional molecular pathways involved in the aggressive behavior of FS-DFSP.	[158]
<i>CDKN2A/B</i> Deletion	- Homozygous deletions of tumour suppressor genes CDKN2A and CDKN2B, often seen in aggressive DFSP variants.	- Loss of cell cycle regulation due to these deletions may promote tumour growth and contribute to imatinib resistance	[156]
<i>FAM118B</i> Overexpression	- Increased expression of FAM118B, involved in Cajal body formation and small nuclear ribonucleoprotein (snRNP) biogenesis.	- Potential biomarker for DFSP. - Potential role in promoting cellular growth and YAP-dependent oncogenesis.	[152]
<i>ZNF286A</i> Overexpression	- Overexpression of ZNF286A, a zinc-finger transcription factor.	- Likely contributes to tumorigenesis through transcriptional regulation of key oncogenic pathways.	[152]
Transcriptomic Alterations	- RNA-seq: Differential expression of collagen-related and cell cycle genes. - Overexpression of PDGF pathway-related genes.	- Highlights potential targets for novel therapeutic strategies, including drugs targeting extracellular matrix remodeling	[152]

sequencing revealed that DFSP is associated with many other genetic alterations, often concerning mutations common in other cases of tumours [156].

DFSP is characterised by abnormal activation of specific signalling pathways that drive tumour growth and development [152, 156]. The most important of these is the platelet-derived growth factor receptor (PDGFR) [118, 156]. The majority of DFSP cases involve a chromosomal translocation, specifically a fusion of the COL1A1 gene with the

PDGFB gene, most commonly involving chromosomes 17 and 22 [163]. This fusion results in the production of a COL1A1-PDGFR β fusion protein, which leads to continuous activation of the PDGFR. The fusion protein acts as an auto-crine factor, binding to the PDGFR on the surface of DFSP cells, which in turn stimulates cell growth and proliferation [164]. The PDGFR has an extracellular domain that binds the growth factor ligand, a transmembrane domain and a cytoplasmic domain with tyrosine kinase activity [165].

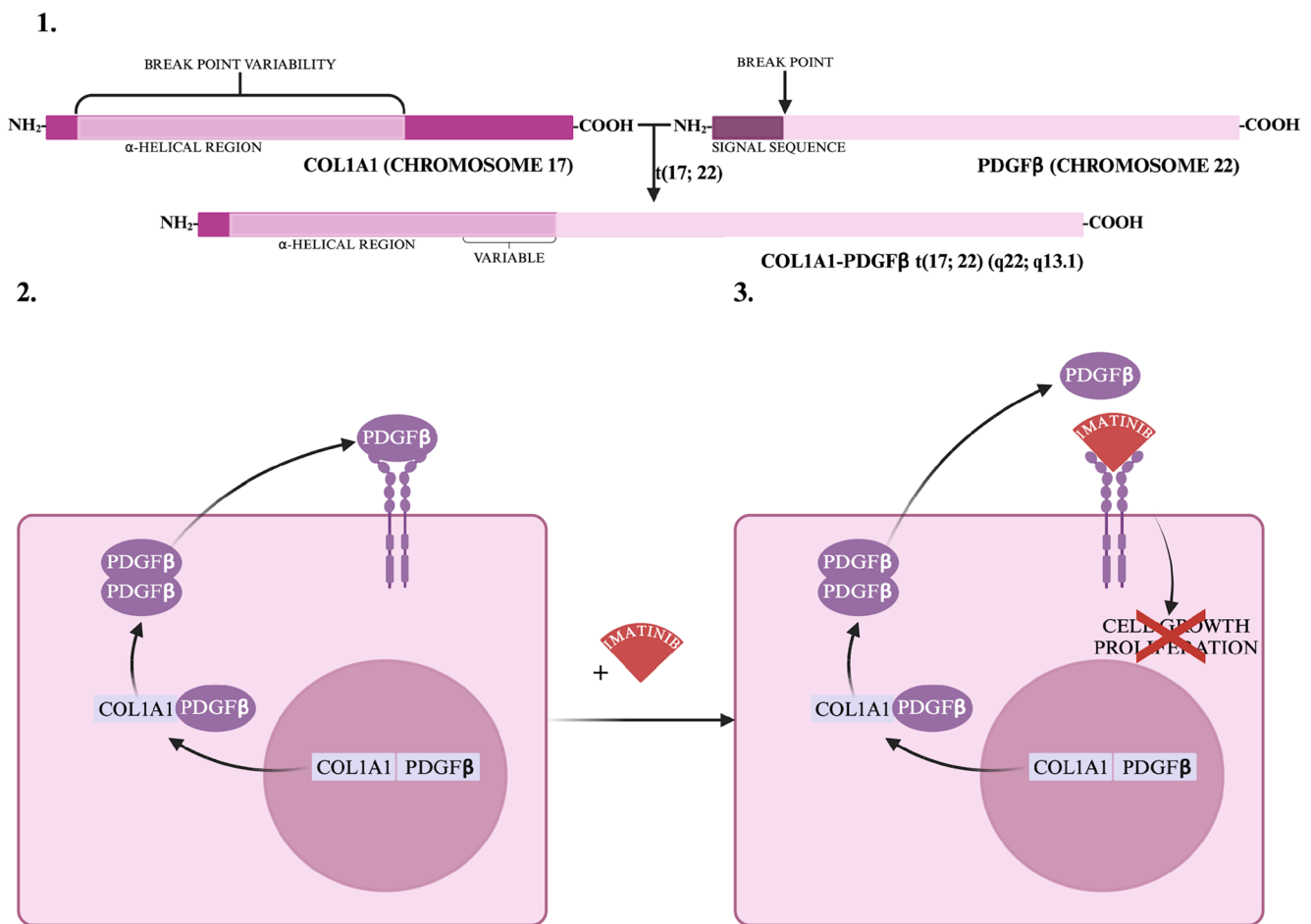


Fig. 3 Molecular mechanism of COL1A1-PDGFB fusion gene and imatinib action in DFSP: (1) formation of COL1A1-PDGFB fusion gene; (2) autocrine PDGFB signaling loop in tumour cells; (3) Imatinib's mechanism of action.

Upon ligand binding, the receptor activates intracellular signalling cascades that trigger processes such as proliferation, chemotaxis and apoptosis. Imatinib, a tyrosine kinase inhibitor, is effective in treating DFSP because it blocks the ATP-binding site of the PDGFR, reducing its enzymatic activity and inhibiting the growth of DFSP cells [166, 167].

In addition to the PDGFR signalling pathway, the fibroblast growth factor receptor (FGFR) signalling pathway also plays a role in DFSP [168]. DFSP cells express FGFRs, in particular FGFR1 and FGFR2, as well as their ligands, fibroblast growth factors (FGFs) [169]. Binding of FGFs to FGFRs leads to autocrine or paracrine activation of the signalling pathway. In autocrine activation, FGFs produced by DFSP cells bind to FGFRs on the same cells, leading to continuous signalling [170]. In paracrine activation, FGFs produced by stromal cells activate FGFRs on DFSP cells [164]. The COL1A1-PDGFB fusion protein can also enhance FGFR signalling. Activation of the FGFR signalling pathway promotes cell proliferation, survival and migration, contributing to tumour growth. Several FGFR inhibitors are being investigated as potential therapies for DFSP.

Activation of the PDGFR initiates several downstream signalling pathways (Table 7). The Ras-MAPK pathway is activated by PDGFR and is important for cell proliferation and differentiation. Its activation by PDGFR signalling contributes to the continuous proliferation of DFSP cells [164]. The PI3K-Akt pathway is also activated by PDGFR and is important for cell survival and resistance to apoptosis, promoting tumour progression and resistance to therapy [174, 179]. Understanding these signalling pathways is crucial for the development of targeted therapies for DFSP.

Conclusions

An accurate diagnosis of DFSP depends on recognising its characteristic histopathology, namely storiform spindle cells with diffuse CD34 positivity, and confirming the pathognomonic *COL1A1-PDGFB* gene fusion via FISH or RT-PCR. This fusion enables the use of targeted therapy with imatinib, particularly in advanced or unresectable cases. Clinicians should remain vigilant for DFSP in the presence of

Table 7 Key signaling pathways in DFSP with targeted therapies under investigation

Signalling pathway	Description	References
PDGF (Platelet-Derived Growth Factor)	-Characteristic COL1A1-PDGFB fusion gene drives DFSP pathogenesis. -Fusion produces a PDGF- β -like protein leading to autocrine activation of PDGFR (especially PDGFRB). -Activated PDGFR triggers downstream pathways that promote cell proliferation, survival, and migration. -Imatinib targets this signalling pathway.	[171–173]
Ras-MAPK (Mitogen-Activated Protein Kinase)	-Activated by PDGFRs; crucial for regulating cell proliferation and differentiation. -Sustained PDGFR signalling drives continuous DFSP cell proliferation and tumour development.	[174–176]
PI3K-Akt (Phosphoinositide 3-Kinase)	-Triggered by PDGFR activation. -Essential for promoting cell survival and resistance to apoptosis. -Contributes to DFSP progression and therapeutic resistance.	[174, 177]
FGFR (Fibroblast Growth Factor Receptor)	-FGFR1 and FGFR2, along with FGFs, are expressed in DFSP cells. -Autocrine activation occurs when DFSP cells secrete FGFs that bind to their own FGFRs, leading to sustained signalling. -Promotes cell proliferation, survival, and migration, ultimately driving tumour growth.	[169, 174]
p53	-Aberrant p53 expression contributes to fibrosarcomatous transformation of DFSP. -p53 mutations and microsatellite instability are linked to tumor progression.	[174, 178]

persistent, slow-growing dermal lesions, particularly those that do not respond to standard treatment. Distinguishing DFSP from benign mimics necessitates the integration of imaging techniques (e.g. MRI and ultrasound), IHC and molecular diagnostics. Identifying FS transformation is crucial because it significantly worsens prognosis, including much higher LR and metastatic rates. Emerging fusion variants (e.g. *COL1A2-PDGFB* and *FBNI-CSAD*) and genetic alterations (e.g. *CDKN2A/B* deletions) may inform future risk stratification and therapeutic resistance. Doctors should prioritise molecular confirmation in cases where diagnosis is uncertain or the tumour is aggressive, and be aware that conventional immunotherapies are not very effective due to the immune-cold microenvironment of DFSP. In clinical practice, early biopsy, molecular testing and multidisciplinary planning are critical for the effective diagnosis and treatment of DFSP.

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Declarations

Competing interests The authors declare no competing interests.

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